

Review and Rating table management

Review table

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Supplier_ID	Foreign Key
Rating	Check (1-5)
Review_Text	-
Review_Date	-

Rating table

Attribute	Key Type
Rating_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	Check (1-5)
Rating_date	date

Create table review

```
SQL> create table review (  
  2     review_id number primary key,  
  3     customer_id number,  
  4     product_id number,  
  5     review_text varchar2(200),  
  6     review_date date  
  7 );
```

Table created.

Create table rating

```
SQL> create table rating (  
  2     rating_id number primary key,  
  3     customer_id number,  
  4     product_id number,  
  5     rating number(1) check (rating between 1 and 5),  
  6     rating_date date  
  7 );
```

Table created.

Insert values of Review table

```
SQL> insert into review values (1, 101, 301, 'good product and quality is nice', '01-jan-2026');
1 row created.

SQL> insert into review values (2, 102, 302, 'very useful product', '02-jan-2026');
1 row created.

SQL> insert into review values (3, 103, 303, 'excellent quality', '03-jan-2026');
1 row created.

SQL> insert into review values (4, 104, 304, 'product is worth the price', '04-jan-2026');
1 row created.

SQL> insert into review values (5, 105, 305, 'average product', '05-jan-2026');
1 row created.

SQL> insert into review values (6, 106, 301, 'really satisfied with the product', '06-jan-2026');
1 row created.

SQL> insert into review values (7, 107, 302, 'good value for money', '07-jan-2026');
1 row created.

SQL> insert into review values (8, 108, 303, 'product quality is excellent', '08-jan-2026');
1 row created.

SQL> insert into review values (9, 109, 304, 'not bad and works well', '09-jan-2026');
1 row created.

SQL> insert into review values (10, 110, 305, 'very happy with this product', '10-jan-2026');
1 row created.

SQL>
SQL> commit;

Commit complete.
```

Insert values into Rating

```
SQL> insert into rating values (1, 101, 301, 5, '01-jan-2026');
1 row created.

SQL> insert into rating values (2, 102, 302, 4, '02-jan-2026');
1 row created.

SQL> insert into rating values (3, 103, 303, 5, '03-jan-2026');
1 row created.

SQL> insert into rating values (4, 104, 304, 4, '04-jan-2026');
1 row created.

SQL> insert into rating values (5, 105, 305, 3, '05-jan-2026');
1 row created.

SQL> insert into rating values (6, 106, 301, 5, '06-jan-2026');
1 row created.

SQL> insert into rating values (7, 107, 302, 4, '07-jan-2026');
1 row created.

SQL> insert into rating values (8, 108, 303, 5, '08-jan-2026');
1 row created.
```

```
SQL> insert into rating values (9, 109, 304, 3, '09-jan-2026');
1 row created.

SQL> insert into rating values (10, 110, 305, 4, '10-jan-2026');
1 row created.

SQL>
SQL> commit;

Commit complete.
```

Display Review table

```
SQL> select*from review;
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
-----
```

```
REVIEW_TEXT
```

```
-----
```

```
REVIEW_DA
```

```
-----
```

```
1 101 301
```

```
good product and quality is nice
```

```
01-JAN-26
```

```
2 102 302
```

```
very useful product
```

```
02-JAN-26
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
-----
```

```
REVIEW_TEXT
```

```
-----
```

```
REVIEW_DA
```

```
-----
```

```
3 103 303
```

```
excellent quality
```

```
03-JAN-26
```

```
4 104 304
```

```
product is worth the price
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
-----
```

```
REVIEW_TEXT
```

```
-----
```

```
REVIEW_DA
```

```
-----
```

```
04-JAN-26
```

```
5 105 305
```

```
average product
```

```
05-JAN-26
```

```
6 106 301
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
-----
```

```
REVIEW_TEXT
```

```
-----
```

```
REVIEW_DA
```

```
-----
```

```
really satisfied with the product
```

```
06-JAN-26
```

```
7 107 302
```

```
good value for money
```

```
07-JAN-26
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
-----
```

```
REVIEW_TEXT
```

```
-----
```

```
REVIEW_DA
```

```
-----
```

```
      8      108      303  
product quality is excellent  
08-JAN-26
```

```
      9      109      304  
not bad and works well  
09-JAN-26
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
-----
```

```
REVIEW_TEXT
```

```
-----
```

```
REVIEW_DA
```

```
-----
```

```
     10     110     305  
very happy with this product  
10-JAN-26
```

```
10 rows selected.
```

Display Review with the Ratings

```
SQL> select r.review_id,
2         r.customer_id,
3         r.product_id,
4         r.review_text,
5         a.rating
6 from review r
7 join rating a
8 on r.customer_id = a.customer_id
9 and r.product_id = a.product_id;
```

REVIEW_ID	CUSTOMER_ID	PRODUCT_ID	REVIEW_TEXT	RATING
1	101	301	good product and quality is nice	5
2	102	302	very useful product	4

REVIEW_ID	CUSTOMER_ID	PRODUCT_ID	REVIEW_TEXT	RATING
3	103	303	excellent quality	5
4	104	304	product is worth the price	

REVIEW_ID	CUSTOMER_ID	PRODUCT_ID	REVIEW_TEXT	RATING
5	105	305	average product	3
6	106	301		

```

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
      RATING
-----
really satisfied with the product
      5

      7      107      302
good value for money
      4

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
      RATING
-----
      8      108      303
product quality is excellent
      5

      9      109      304
not bad and works well
      3

```

```

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
      RATING
-----

      10      110      305
very happy with this product
      4

10 rows selected.

```

Calculate Average product ratings

```
SQL> select product_id,  
2         avg(rating) as average_rating  
3   from rating  
4  group by product_id;
```

PRODUCT_ID	AVERAGE_RATING
301	5
302	4
303	5
304	3.5
305	3.5

Identify Highly rated products

```
SQL> select product_id,  
2         avg(rating) as average_rating  
3   from rating  
4  group by product_id  
5  having avg(rating) >= 4;
```

PRODUCT_ID	AVERAGE_RATING
301	5
302	4
303	5

Display the highest rated product

```
SQL> select product_id,
2         avg(rating) as average_rating
3   from rating
4  group by product_id
5  order by avg(rating) desc;
```

PRODUCT_ID	AVERAGE_RATING
301	5
303	5
302	4
305	3.5
304	3.5

Count the number of reviews for each product

```
SQL> select product_id,
2         count(*) as total_reviews
3   from review
4  group by product_id;
```

PRODUCT_ID	TOTAL_REVIEWS
301	2
302	2
303	2
304	2
305	2

Display products with 5-star ratings

```
SQL> select product_id,
2         customer_id,
3         rating
4   from rating
5  where rating = 5;
```

PRODUCT_ID	CUSTOMER_ID	RATING
301	101	5
303	103	5
301	106	5
303	108	5